

Year 10 Engineering Technology - Water Control System

Term 1 lesson plans: 30 lessons over 10 weeks

Lesson Plan Overview

These lesson plans support a 10-week Stage 5 Engineering Technology Control systems unit. They assume three lessons per week, matching the Year 9 Term 1 lesson-plan structure. Outcome alignment uses the NESA Stage 5 EGT5 outcomes and the Control systems focus area.

Primary Outcomes

- EGT5-SAF-01: applies risk management and safe work practices in engineering contexts
- EGT5-COM-01: communicates ideas, concepts and solutions for engineering practice
- EGT5-EVL-01: investigates and evaluates engineering systems and solutions
- EGT5-IVT-01: explains engineering practices and the influence of technologies used in engineering industries
- EGT5-GRP-01: develops and applies technical graphics in engineering contexts
- EGT5-MEA-01: applies mechanical analysis and practical testing to investigate engineering concepts
- EGT5-USE-01: selects and applies materials for engineering projects
- EGT5-ENV-01: analyses relationships between engineering design, production and sustainability

Lesson Sequence

Week 1 - Launch, WHS and Evidence Setup

Lesson 1 - Unit Introduction and Water-Control Context (Classroom)

Learning intention

- Understand the Water Control System project brief and why automated watering systems are engineered.

Success criteria

- I can describe the project problem and user need.
- I can explain why control systems are used in irrigation.

Activities

- Introduce the Water Control System project brief, constraints and folio expectations.
- Show examples of automatic irrigation, smart garden systems and industrial water-control systems.
- Class discussion: where does automation improve efficiency, safety or sustainability?
- Students write a first project brief response and three success criteria.

Resources

- Year 10 Term 1 website page.
- System overview poster.
- Sample irrigation/control-system images.

Evidence

- Project brief response.
- Student success criteria.

Lesson 2 - WHS: Water, Electricity and Safe Testing (Workshop/Classroom)

Learning intention

- Apply risk management and safe work practices to low-voltage water-control testing.

Success criteria

- I can identify hazards in a water-control project.
- I can propose controls using wet/dry zones and teacher check points.

Activities

- Review low-voltage expectations and the no-mains-power rule.
- Demonstrate wet zone, dry electronics zone, spill response and power-off adjustment routine.
- Students complete an initial risk-control table.
- Teacher checks understanding using short safety scenarios.

Resources

- Safe test setup poster.
- WHS checklist.
- Risk-control template.
- PPE.

Evidence

- Risk-control table.
- Safe setup sketch.

Lesson 3 - Workshop Expectations and Evidence Setup (Workshop/Classroom)

Learning intention

- Set up the expectations, evidence habits and folio structure for the project.

Success criteria

- I can explain workshop expectations for this project.
- I can identify the evidence I need to collect as I work.

Activities

- Review behaviour, tool-use, PPE and clean-up expectations.
- Walk through the online portfolio workbook and evidence checklist.
- Students create or open their folio file and set up section headings.
- Start a project glossary: input, process, output, feedback, sensor, actuator.

Resources

- Portfolio workbook.
- Testing checklist.
- Website toolkits.
- Student device or workbook.

Evidence

- Folio file started.
- Glossary started.
- Teacher observation.

Week 2 - Control-System Concepts

Lesson 4 - Input, Control and Output (Classroom)

Learning intention

- Use correct terminology to explain the parts of a control system.

Success criteria

- I can label input, control/process, output and feedback.
- I can explain one example of each part in a watering system.

Activities

- Direct instruction: input, control/process, output, actuator, sensor and feedback.
- Students annotate the system overview diagram.
- Pairs compare manual watering, timed watering and sensor-triggered watering.
- Exit ticket: explain one input and one output.

Resources

- System overview poster.
- Glossary.
- Mini whiteboards or notebooks.

Evidence

- Annotated IPO diagram.
- Exit ticket.

Lesson 5 - Open-loop and Closed-loop Control (Classroom)

Learning intention

- Explain the difference between open-loop and closed-loop control in a watering context.

Success criteria

- I can compare a timer-only system with a sensor-feedback system.
- I can explain why feedback can improve water efficiency.

Activities

- Teacher models open-loop and closed-loop examples.
- Students sort system examples into open-loop or closed-loop.
- Students write a short paragraph explaining which is better for water efficiency and why.
- Class shares strongest comparison statements.

Resources

- Example cards.
- Flowchart poster.
- Student notebooks.

Evidence

- Open/closed-loop sort.
- Comparison paragraph.

Lesson 6 - Control-System Vocabulary and Block Diagrams (Classroom)

Learning intention

- Communicate control-system relationships using block diagrams and subject terminology.

Success criteria

- I can create a block diagram for a watering system.
- I can use key control-system terms accurately.

Activities

- Model a block diagram: input -> controller -> output -> feedback.
- Students convert a written system description into a block diagram.
- Vocabulary check: threshold, actuator, controller, relay, feedback, reliability.
- Students add a cleaned-up block diagram to the folio.

Resources

- Block diagram examples.
- System overview poster.
- Portfolio workbook.

Evidence

- Block diagram.
- Vocabulary notes.

Week 3 - Components, Electricity and Water Path

Lesson 7 - Electronic and Hydraulic Components (Classroom/Workshop)

Learning intention

- Investigate input and output components used in electronic and hydraulic control systems.

Success criteria

- I can explain the purpose of a sensor/switch, controller/relay, pump/valve, reservoir and tubing.

Activities

- Component handling stations: sensors, switches, relays/controllers, pumps/valves and tubing.
- Students complete a component purpose table.
- Teacher demonstration of a safe pump/valve output test without full system connection.
- Students identify which components are input, control or output.

Resources

- Component kit.
- Sensors/actuators poster.
- Sample pump/valve setup.

Evidence

- Component table.
- Teacher observation.

Lesson 8 - Basic Electrical Testing (Workshop/Classroom)

Learning intention

- Use simple testing ideas to check low-voltage circuits safely.

Success criteria

- I can explain why circuits are checked before power.
- I can describe voltage, current and resistance at a basic level.

Activities

- Teacher explains voltage, current and resistance in project-friendly language.
- Demonstrate checking polarity, loose connections and power-off adjustments.
- Students practise identifying safe and unsafe wiring examples.
- Optional teacher-led multimeter demonstration.

Resources

- Low-voltage circuit samples.
- Multimeter for demonstration.
- Safe wiring examples.

Evidence

- Wiring safety notes.
- Safe/unsafe example responses.

Lesson 9 - Hydraulic Water Path Principles (Classroom/Workshop)

Learning intention

- Understand how reservoir, pump, valve, tubing and outlet choices affect water movement.

Success criteria

- I can explain how water moves through the system.
- I can identify likely leak, kink or blockage points.

Activities

- Demonstrate reservoir, tubing, pump/valve and outlet arrangement.
- Students sketch a simple water path and label flow direction.
- Discuss pressure, flow, leaks, kinks and blockages in practical terms.
- Students add water path risks to their risk table.

Resources

- Reservoir, pump/valve and tubing.
- Safe test tray.
- Water path poster.

Evidence

- Annotated water path sketch.
- Updated risk table.

Week 4 - Materials, Concepts and Design Justification

Lesson 10 - Materials and Component Selection (Classroom)

Learning intention

- Select components and materials for safety, reliability, durability and water efficiency.

Success criteria

- I can justify why particular components and materials suit my design.
- I can consider reuse, recycling and repairability.

Activities

- Discuss tubing, reservoirs, bases, wiring, connectors and plant-bed materials.
- Students compare possible materials/components against criteria.
- Start materials and component list for the folio.
- Teacher checks that choices match the project constraints.

Resources

- Component samples.
- Materials table.
- Portfolio workbook.

Evidence

- Materials/component list with justification.

Lesson 11 - Concept Sketching (Classroom)**Learning intention**

- Generate and communicate multiple design ideas for a water-control system.

Success criteria

- I can produce at least two annotated concept sketches.
- I can show water path, components and safety zones.

Activities

- Review sketch expectations and annotation examples.
- Students sketch 2-3 possible system layouts.
- Peer feedback: clarity, safety, testability and water path.
- Students select the strongest concept and explain why.

Resources

- Design planning poster.
- Sketch paper.
- Rulers.
- Portfolio workbook.

Evidence

- Concept sketches.
- Peer feedback notes.
- Chosen concept justification.

Lesson 12 - Success Criteria and Design Justification (Classroom)**Learning intention**

- Develop clear success criteria and justify design decisions.

Success criteria

- I can write measurable success criteria.

- I can justify design choices using safety, reliability and efficiency.

Activities

- Teacher models weak vs strong success criteria.
- Students write or revise 4-6 criteria for their prototype.
- Students justify chosen input, output and water-path layout.
- Add success criteria and justification to the folio.

Resources

- Folio workbook.
- Design planning poster.
- Examples of success criteria.

Evidence

- Success criteria.
- Design justification paragraph.

Week 5 - Final Design, Algorithm and Approval

Lesson 13 - Final Design Drawing (Classroom)

Learning intention

- Develop a final design drawing that communicates the system clearly.

Success criteria

- I can draw a final layout with labels.
- I can show component positions, water path and wet/dry zones.

Activities

- Students produce final design drawing from chosen concept.
- Add labels for reservoir, tubing, outlet, input, controller, output and power.
- Teacher checks for buildability and safety.
- Students revise drawing before approval.

Resources

- Sketching equipment.
- Design planning poster.
- Portfolio workbook.

Evidence

- Final design drawing.

Lesson 14 - Algorithm or Flowchart (Classroom/ICT)

Learning intention

- Represent control logic using a flowchart or pseudocode.

Success criteria

- I can write a clear algorithm with trigger, action, stop condition and fault response.

Activities

- Analyse a simple watering algorithm.

- Students create their own flowchart or pseudocode.
- Teacher conference checks logic before build.
- Students add final algorithm to folio.

Resources

- Algorithm flowchart poster.
- Flowchart template.
- Portfolio workbook.

Evidence

- Flowchart or pseudocode draft.

Lesson 15 - Build Planning and Teacher Approval (Workshop/Classroom)

Learning intention

- Prepare a safe build sequence and receive teacher approval before construction.

Success criteria

- I can plan build order, testing order and risk controls before starting practical work.

Activities

- Students complete build sequence and test order.
- Teacher reviews design, risk controls, component list and first test plan.
- Students revise plans before build starts.
- Record teacher approval in folio.

Resources

- Testing checklist.
- Portfolio workbook.
- Approved component kit.

Evidence

- Approved plan.
- Teacher check record.

Week 6 - Build Water Path and Output Subsystem

Lesson 16 - Build the Base and Water Path (Workshop)

Learning intention

- Construct the base, reservoir, tubing and outlet safely.

Success criteria

- I can build the water path and check that it is secure before adding electronics.

Activities

- Set up base, reservoir, tubing and plant-bed area.
- Secure tubing and check for kinks or loose joins.
- Leak test with power disconnected.
- Record build photos and notes in folio.

Resources

- Base board, reservoir, tubing, fittings, plant-bed tray, towels/spill kit.

Evidence

- Build log photos.
- Leak-test notes.

Lesson 17 - Test the Output Subsystem (Workshop)

Learning intention

- Test the water output subsystem safely and collect basic data.

Success criteria

- I can test a pump/valve output and record water delivered over time.

Activities

- Teacher demonstrates output-only testing.
- Students test pump/valve under supervision.
- Record water delivered over a fixed time and any leaks/faults.
- Update build log and risk controls if needed.

Resources

- Pump or valve, low-voltage supply, measuring cylinder, timer, spill kit.

Evidence

- Output test data.
- Build log update.

Lesson 18 - Improve the Water Path (Workshop)

Learning intention

- Use test evidence to improve the water path before full-system integration.

Success criteria

- I can identify one water-path issue and make a justified improvement.

Activities

- Students review output test results.
- Identify leaks, weak flow, kinks, poor outlet aim or insecure mounting.
- Modify one issue at a time.
- Retest and record whether the change improved performance.

Resources

- Prototype, tubing/fittings, test table, towels/spill kit.

Evidence

- Retest result.
- Fault/improvement note.

Week 7 - Build Input and Control Subsystem

Lesson 19 - Build the Input Subsystem (Workshop)

Learning intention

- Construct and test the input subsystem.

Success criteria

- I can test a switch or sensor and identify whether it changes state correctly.

Activities

- Set up input device and simple test arrangement.
- Teacher checks setup before power.
- Students test trigger response and record issues.
- Update input notes in folio.

Resources

- Switch or sensor, controller/relay as required, low-voltage power.

Evidence

- Input test data.
- Teacher observation.

Lesson 20 - Build the Control Subsystem (Workshop)

Learning intention

- Connect the controller/relay/circuit safely and explain its role in the system.

Success criteria

- I can explain how the controller/process links input to output.
- I can follow teacher check procedures before powered testing.

Activities

- Teacher models controller/relay connections.
- Students wire the control subsystem with power disconnected.
- Teacher check before power.
- Students test trigger-to-output behaviour using a safe short test.

Resources

- Controller/relay/circuit, low-voltage power, wiring, teacher check sheet.

Evidence

- Wiring photo/diagram.
- Teacher check record.

Lesson 21 - Update Wiring Diagram and Safety Evidence (Classroom/Workshop)

Learning intention

- Document the actual wiring/control setup clearly after build changes.

Success criteria

- I can update a wiring/control diagram so it matches my prototype.
- I can explain how safety controls are applied.

Activities

- Students compare original plan with actual prototype.
- Update wiring/control diagram and label wet/dry zones.

- Add photo evidence and a short safety reflection.
- Teacher checks folio evidence before full integration.

Resources

- Prototype, folio workbook, camera/device, checklist.

Evidence

- Updated diagram.
- Safety reflection.

Week 8 - Integration, Troubleshooting and Testing

Lesson 22 - Integrate the Full System (Workshop)

Learning intention

- Combine subsystems into a working water-control prototype.

Success criteria

- I can connect input, control and output safely.
- I can describe how the full system should behave.

Activities

- Connect input, controller and output after teacher approval.
- Run short tests first.
- Power off before adjustments.
- Record whether the system responds as intended.

Resources

- Approved prototype parts, testing checklist, spill kit.

Evidence

- Integrated system test notes.

Lesson 23 - Troubleshooting and Fault Log (Workshop)

Learning intention

- Document troubleshooting processes and solutions clearly.

Success criteria

- I can record faults, likely causes, fixes and retest results.

Activities

- Teach fault-log structure.
- Students identify one fault or improvement target.
- Modify one variable at a time and retest.
- Update fault log with evidence.

Resources

- Fault log template.
- Testing checklist.
- Prototype.

Evidence

- Fault log.
- Retest data.

Lesson 24 - Repeated Testing and Efficiency (Workshop)

Learning intention

- Test the control system to determine reliability and efficiency.

Success criteria

- I can use repeated trials to judge whether my system is reliable and efficient.

Activities

- Run repeated full-system test cycles.
- Record trigger condition, response time, water delivered, leaks/faults and result.
- Compare consistency across trials.
- Discuss what the data shows.

Resources

- Test table, measuring cylinder, timer, prototype.

Evidence

- Completed full-system test table.

Week 9 - Analysis, Sustainability and Report Drafting

Lesson 25 - Data Presentation and Analysis (Classroom/ICT)

Learning intention

- Communicate test results using tables, graphs and written explanation.

Success criteria

- I can present test data and explain what it shows about system performance.

Activities

- Students tidy test table and create a simple graph or summary.
- Teacher models evidence-based evaluation sentences.
- Students write results analysis paragraph.
- Folio workshop: check evidence gaps.

Resources

- Folio workbook.
- Spreadsheet or graph paper.
- Test data.

Evidence

- Data presentation.
- Analysis paragraph.

Lesson 26 - Sustainability and Improvement (Classroom)

Learning intention

- Evaluate environmental impact, water efficiency and material/component choices.

Success criteria

- I can explain how the system could save water, reduce waste or be improved.

Activities

- Discuss water efficiency, materials, reuse/recycling and repairability.
- Students identify two sustainability strengths and two improvements.
- Add sustainability reflection to folio.
- Link improvements back to evidence from testing.

Resources

- Sustainability prompts.
- Materials notes.
- Test data.

Evidence

- Sustainability reflection.

Lesson 27 - Engineering Report Drafting (Classroom/ICT)**Learning intention**

- Develop an engineering report using appropriate structure and terminology.

Success criteria

- I can organise my folio/report with brief, design, build, testing and evaluation sections.

Activities

- Review report/folio checklist.
- Students assemble evidence into the portfolio workbook.
- Teacher checks missing evidence and gives targeted feedback.
- Students write or revise design and build sections.

Resources

- Portfolio workbook.
- Assessment checklist.
- Student evidence.

Evidence

- Draft folio/report.

Week 10 - Wrap, Reflection and Submission**Lesson 28 - Safety, Environment and Industry Links (Classroom)****Learning intention**

- Understand how safety, environment and industry practice connect to real engineering work.

Success criteria

- I can explain one safety issue, one environmental issue and one industry connection related to control systems.

Activities

- Discuss real-world irrigation, water management, IoT, automation and maintenance.
- Students connect their prototype to one industry application.

- Revise safety and sustainability sections of folio.
- Optional small group share-out.

Resources

- Industry examples.
- Syllabus/website notes.
- Folio workbook.

Evidence

- Industry link paragraph.
- Updated safety/sustainability notes.

Lesson 29 - Peer Review and Final Evaluation (Classroom)

Learning intention

- Use peer review to improve communication and final evaluation quality.

Success criteria

- I can give and receive useful feedback.
- I can improve my evaluation using test evidence.

Activities

- Peer review using criteria: clarity, evidence, terminology, safety and sustainability.
- Students revise final evaluation.
- Teacher conferences with students needing support.
- Final missing-evidence check.

Resources

- Peer review checklist.
- Portfolio workbook.
- Assessment criteria.

Evidence

- Peer feedback.
- Revised evaluation.

Lesson 30 - Submission and Project Reflection (Classroom/Workshop)

Learning intention

- Present project outcomes and reflect on engineering learning.

Success criteria

- I can explain how my system works, what the data showed and what I would improve.

Activities

- Final submission check.
- Short informal presentations or gallery walk.
- Final reflection: design decision, fault solved, improvement for next time.
- Collect prototype/folio evidence.

Resources

- Completed prototype.

- Completed folio.
- Reflection prompt.

Evidence

- Submitted folio/prototype.
- Final reflection.